Google Developer Group
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Webinar

Prompt Engineering & Context Engineering

Get better results from AI agents by giving clear instructions and the right context.



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Webinar

Today's Session

Understand AI agents and learn how to get better results from them.

What we'll cover

- AI agents and how they work
- **Prompt engineering** (better instructions)
- **Context engineering** (the right information)

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Chatbot vs AI Agent

Two ways to work with AI, with different capabilities and goals.

Chatbot

A conversational AI that responds to your messages.

Question → Response

AI Agent

A goal-driven AI that can plan, use tools, and take actions to get things done.

Goal → Tools → Result

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How AI Agents Work

From input to action — under the hood.

The Agent Loop



LLM
The reasoning engine.

Tokens
Pieces of language (the model's language).

Reasoning
Multi-step problem solving.

Memory
Context, history, and retrievel.

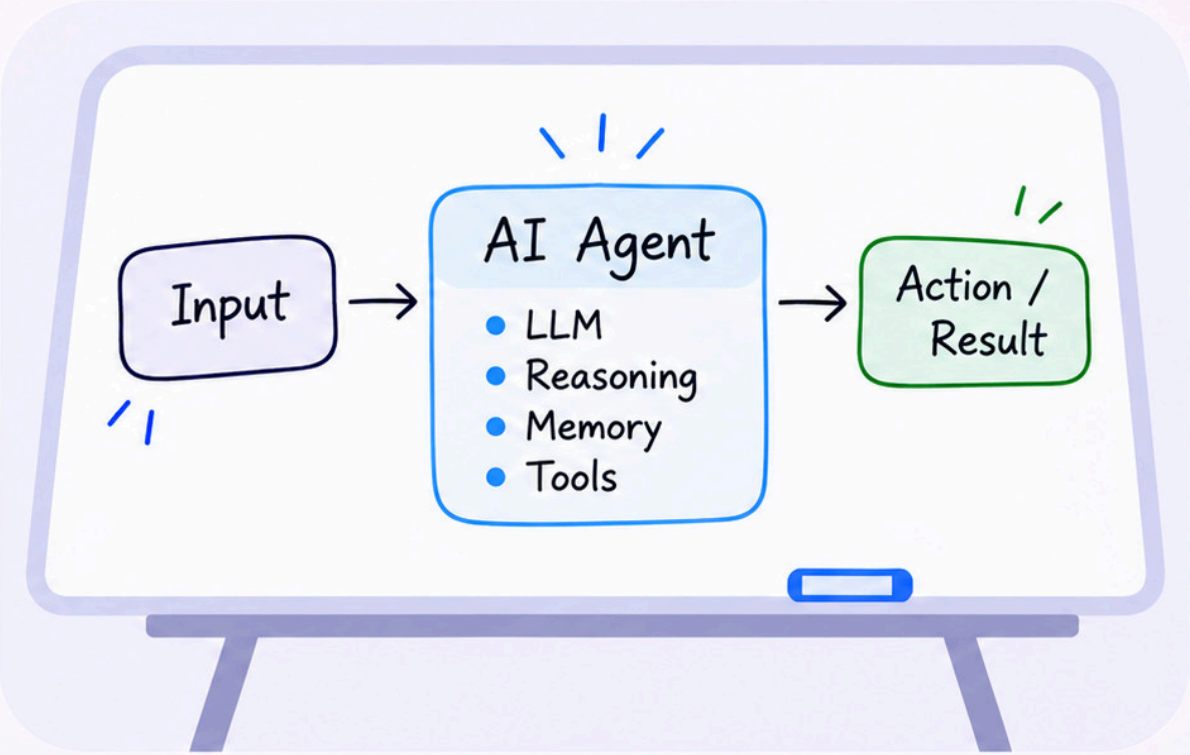
Tools
Interact with external systems.

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How AI Agents Work

How AI Agents Work

Let's break this down together on the whiteboard and see what happens inside an AI agent.



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Key Takeaways

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The big ideas to remember about AI agents.



More than chatbots

They can reason, plan and take action.



Built from key building blocks

LLMs, tokens, reasoning, memory and tools.



Turn ideas into action

Automate tasks and solve real problems.

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Prompt Engineering

Communicating with AI

How do we give clear instructions and get better results?



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Prompt Engineering

What is Prompt Engineering?

The skill of writing clear, specific instructions to get better results from AI.



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Prompt Engineering

Good vs. Bad Prompts

See how the way you ask changes the quality of the result.

Bad Prompt

Write a blog post about AI.

Good Prompt

Write a 1,000-word beginner-friendly blog post about AI agents, with examples, real-world uses, and a clear conclusion.

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Prompt Engineering

The 5 Parts of a Good Prompt

A simple structure to write clear, specific, and effective prompts for better results.

Role
Tell the AI who to act as (e.g. a tutor, developer).

Task
State what you want it to do (e.g. write, explain, compare).

Context
Give the background and relevant details.

Constraints
Set limits (e.g. length, format, tone, tools).

Desired Output
State the exact result you want (e.g. a blog post, a list).

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Prompt Engineering

Advanced Prompt Engineering

Practical techniques to get more accurate, useful, and reliable results.

Task decomposition
Break complex tasks into smaller, clearer steps.

Few-shot prompting
Give examples of the expected input and output.

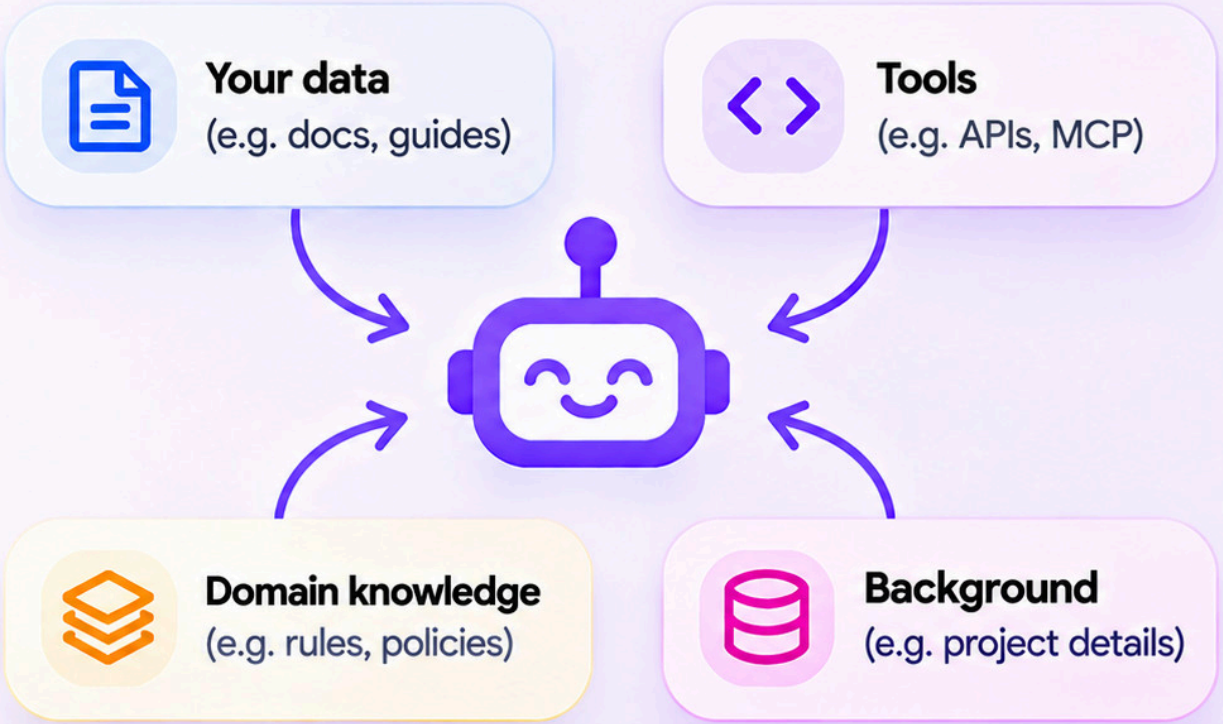
Iterate & refine
Review the result, give feedback, and improve it.

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Context Engineering

Context Engineering

Give AI the right information, resources, and environment to do its best work.



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Prompt Engineering

Prompt vs. Context Engineering

Both work together to help AI give better, more useful results.

Prompt Engineering

The instructions.
You tell AI what to do.

Example
"Summarize this document for a beginner."

+

Context Engineering

The supporting information.
You give AI the right background, data, and tools.

Example
"Here's the document, the target audience, and the key points to cover."

Together: clear instructions + the right context = higher quality results.

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Live Demo

Let's Build Better AI Workflows

From idea to a clear plan and execution, with Gemini and Antigravity.

Gemini — Plan & Design

Refine the idea
Turn a raw idea into a clear, viable plan.

Write a system prompt
Define the AI's role, behavior, and guardrails.

Generate a PRD
Create a clear product requirements doc.

Act as the AI agent builder
Break the work into phases and implementation milestones.

Antigravity — Build & Ship

Turn the plan into code
Build features step by step.

Follow the milestones
Ship in phases with clear progress.

Get a working result
A usable, testable app or feature.

Demo: Idea → Plan (PRD + milestones) → Build in Antigravity

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Key Takeaways

Key Takeaways

The main lessons from today's session.

Prompt engineering
Gives clear, effective instructions to AI.

Context engineering
Provides the right information, tools, and environment.

Better workflows
Combine both to solve real problems end to end.

AI as a builder
Use tools like Gemini and Antigravity to go from idea to a working result.

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What's Next?

Agent Skills

Turn what we've learned into reusable, powerful capabilities for your AI agents.



Reusable capabilities
Package instructions, context, tools, and workflows.

Run on demand
Give your agents new abilities (e.g. plan, write, build, research).

Build and share
Create skills for your own projects and team workflows.

Next Session: Building and Using Agent Skills →

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Thank You!

Questions, thoughts, or discussions?



Keep building
Apply today's ideas to a real task.

Share your progress
in the Web Dev group.

See you at the next session
Building and Using Agent Skills.

Build • Experiment • Share • Grow